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Two-Tier Device-Based Authentication Protocol Against PUEA Attacks for IoT Applications

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***Abstract*—Securing internet of things (IoT) applications is complicated by the limited power and processing capabilities of a typical sensor node. This paper proposes a protocol and a method of spectrum management that can guard against common types of security threats despite the limitations of the local processing. With a hierarchical cognitive IoT architecture, this work incorporates the strengths of cognitive radio and the physical properties of a device to improve the performance of a cognitive radio sensor network (CRSN) and resolve the security problem, considering one of the most destructive attacks in CRSNs called the primary user emulation attack (PUEA). Accordingly, for delivering trustable services, we explore the trade-off between spectrum management and the successful detection rate of malicious nodes. The proposed distributed secure algorithm allows the sensing sensors and the tasking nodes to perform a detection and identification mechanism such that dynamic spectrum management and correct spectrum decision can be achieved. The experimental results show that the proposed secure system provides a significant protection against the PUEA attacks.**

***Index Terms*—primary user emulation attack, cognitive radio sensor networks, Internet of Things, node identification.**

I. Introduction

Since current wireless systems have a large number of vulnerabilities, more effective defense schemes must be considered. In this situation, for a sensor-driven system, the traditional Internet architecture and protocols need to be augmented to match the IoT service requirements. Due to the emerging demand for Internet of Things (IoT) applications, the use of cognitive computing technologies in combination with data generated by connected devices may play a role in the core of system design [1]. Therefore, cognitive IoT may become the next leap in improving the accuracy and efficiency of complex, sensor-driven systems through learning and infusing more sensing awareness into the surrounding environments. However, due to the characteristics (e.g., cognitive capability and reconfigurability) of a cognitive radio sensor network (CRSN), CRSNs may be vulnerable to a number of novel attacks during the cognitive cycle [2]. Moreover, as mentioned in [3], managing identity and authentication in the IoT is an important issue such that trustable services can be provided. This work considers some of the inherent features of the IoT and explores promising approaches that could be used to provide identification and authentication in a connected hierarchical IoT environment.

Studies [4][5] show that the sensor nodes use unlicensed bands, such as industrial, scientific, and medical (ISM) bands, which are extremely crowded and may yield significant challenges for achieving reliable communication due to packet losses, collisions, and contention delay. Therefore, opportunistic spectrum access of cognitive radio sensor networks may help mitigate these challenges such that critical information requiring real-time reliable communication may exploit the potential advantages of dynamic spectrum management. However, it is noted that the denial-of-service (DoS) attacks could lead to big disadvantages because several good users could lose access to the network due to the presence of a few malicious users, which implies that a bad spectrum decision may lead to poor usage of spectrum for authorized users and an unfair advantage for the malicious users. Therefore, effective secure operations should be carried out in order to obtain awareness about the spectrum usage and the possible presence of primary users. Accordingly, this paper considers one of the most destructive DoS attacks in CRSNs called the primary user emulation attack (PUEA), which mimics a primary user (PU) by transmitting fake signals [6], [7], leading other good secondary users to vacate the spectrum unnecessarily. Therefore, considering the PUEA attack, the *Two-Tier Device-Based Authentication Protocol* ($\mathrm{T^2DAP}$) is proposed to carry out effective secure operations in order to obtain awareness about the spectrum usage and the possible presence of primary users.

In order to maintain network operation and defend against PUEA attacks, a fully distributed secure method is developed from the perspectives of spectrum management with separate sensing and from the perspective of device-based node identification. The first selects sensing sensors [8] and develops a detection and notification mechanism, considering information exchange and the handshaking protocol between the sensing sensors and the tasking nodes (i.e. cognitive nodes). The second perspective relies on physical properties of the devices (e.g., the internal component delays, the difference between the clocks, and the response delays) to identify the malicious

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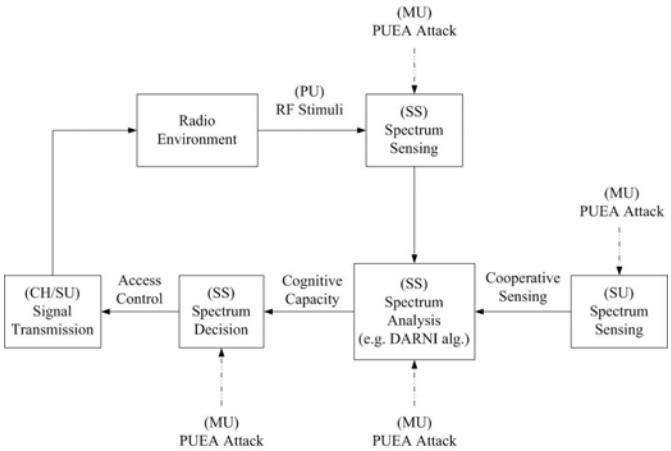

Fig. 1. The cognitive cycle against PUEA attacks (modified and reproduced from [2]).

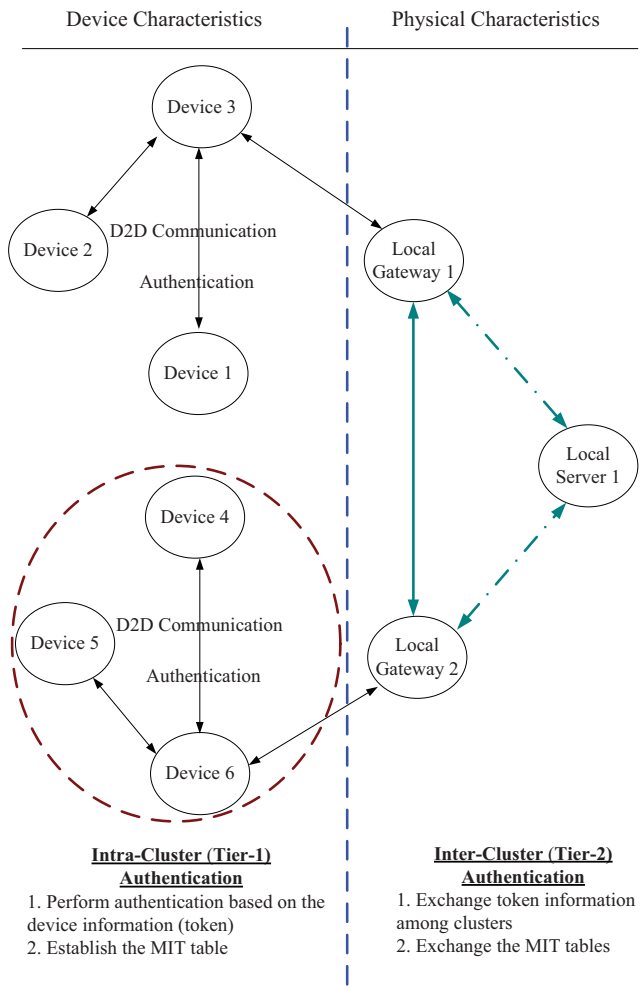

Fig. 2. Preventing PUEA attacks using a two-fold strategy depending on whether the communication is between clusters or within clusters. Details of the proposed defenses are outlined above and details are provided in Figs 6 and 7 (Section III).

nodes. Therefore, a tasking group in a hierarchical CRSN may consist of (1) sensing sensors (SSs) for spectrum sensing and analysis and (2) cognitive nodes (e.g., the clusterheads and cluster members) for sensor tasking. Denote the target (e.g., patients or habitat monitoring) and the cognitive node as the primary user (PU) and the secondary user (SU), respectively.

In contrast to traditional sensing systems, which mainly focus on sensing information collection, data processing, and network management, this work aims to develop systems that further exploit the capabilities of sensing devices (e.g. device characteristics) and extend the capabilities to security issues in IoT applications. This work applies the physical properties of a device, the communication delay (e.g., internal component delay, difference of the clocks, and the response delay) between two wireless sensors such that the malicious nodes which mimic the PU can be identified. The conceptual cognitive cycle against PUEA attacks is shown in Figure 1, including the operations of spectrum sensing, spectrum analysis, and spectrum decision. The proposed distributed secure algorithm with the knowledge of separate sensing allows the sensing sensors and the tasking nodes to perform a detection and identification mechanism such that dynamic spectrum management and correct spectrum decision can be achieved.

To illustrate the concept of this work, a possible IoT healthcare scenario is considered. Note that one way to perform authentication in a network of systems is through certain system information, say a security token in a ward. In this scenario, how to make a reliable authentication decision is a key issue. This work aims to apply the device characteristics to identify the devices. Thus, in contrast to the conventional hardware/software authentication solutions, given the development of device information with the database or reference repositories, new security services may be effectively provided by applying the device information, regarding device characteristics and capabilities. Figure 2 shows a two-fold authentication strategy, which suggests that the proposed method may be a sensible way to identify the nodes by using the device characteristics (e.g., device delay, a variable message length or time asynchronization) in the intra-net of things. Similarly, considering the physical characteristics (e.g., using a response delay time), we aim to develop feasible schemes for node identification in the heterogeneous internet of things. In this work, the delay is at the level of logical gates, ranging from more than 10 ns down to the picosecond range [9].

This paper is organized as follows: In Section II, we review related works about PUEA attacks and spectrum sensing in CRSNs. Section III presents a device-based secure scheme against PUEA attacks. Section IV describes the performance analysis of the proposed scheme. Section V evaluates the system performance and present the performance comparison among the proposed T^2DAP, the belief propagation-based scheme [10] and energy detection-based scheme [11]. Finally, Section VI draws conclusions and shows future research directions.

II. Literature Review

Primary user emulation attack (PUEA) in cognitive radio networks was studied in [6],[12]-[16]. In [6], Chen *et al*

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discuss defense against PUEA by localization of primary transmitters using the angle of arrival, the time of arrival, and the received signal strength of the primary signal. In [12], Chen and Park propose two mechanisms to detect a PUEA, which use the ratio and the difference of the distances of the primary and malicious transmitters from the secondary user to detect a PUEA. In [13], the authors use a hypothesis testing method to detect a primary transmission with spectrum sensing. In [14], security challenges and mitigation technologies are described with respect to the protocol layers. Cooperative secure schemes against PUEA are studied in [17], [18], [19]. In [18], in order to increase resilience to a PUEA, a distributed spectrum decision method is proposed in which SUs make individual spectrum decisions and then exchange individual sensing results with their one-hop neighbors.

Regarding the detection of the PUEAs, received power measurements of the SUs can be applied to detect the attack as well [10],[11],[19]-[22]. In [19], based on the Neyman-Pearson criterion with the aim of maximizing the channel throughput, an optimal strategy is driven for cooperative energy detection and spectrum sensing. In [20], a correlation-based trust factor is defined to identify physical layer attacks in CRNs. In [21], a cross-layer design is proposed, where the channel-tap power is utilized as a radio-frequency fingerprint (RF) to identify PUEAs over multipath Rayleigh fading channels. Note that the above studies assume that the locations of the primary transmitters or the received signal strength of the primary signal are known a priori. In non-location-based algorithms, a helper node (HN) or with cryptographic signatures by HN's signals [23], a public key cryptography mechanism [24], or a AES-assisted scheme [25], [26] can be used to verify the integrated signatures in order to characterize the signal. Instead of just considering energy consumption minimization and spectrum sensing performance, our previous work [27] considers node authentication for intra-cluster communication and tackles the sensing node selection problem and the malicious node identification problem.

Notice that the above works focus on the detection of the PU's activity and PUEA attacks, not carefully considering the impacts of network topology and the collaboration between the cognitive nodes (SUs) on the system performance. In contrast, this work applies time asynchronization and difference of device delay between a MU and the PU for node authentication and the proposed T^2DAP scheme handles the intra-cluster communication and inter-cluster communication with device-based member authentication and device-based gateway authentication, respectively, such that the tradeoffs among the energy efficiency, spectrum sensing performance, and the impact of malicious node behaviors on system performance are clarified.

III. Two-Tier Device-Based Authentication Protocol (T^2DAP)

This section describes the proposed distributed secure method for cooperative spectrum control against PUEA attacks in a cluster-based network topology, which is based on the framework of our previous works [28], [29], which focuses on node identification between a pair of nodes via device information. The characteristics of SSs and the tasking nodes are applied to achieve spectrum management and node identification.

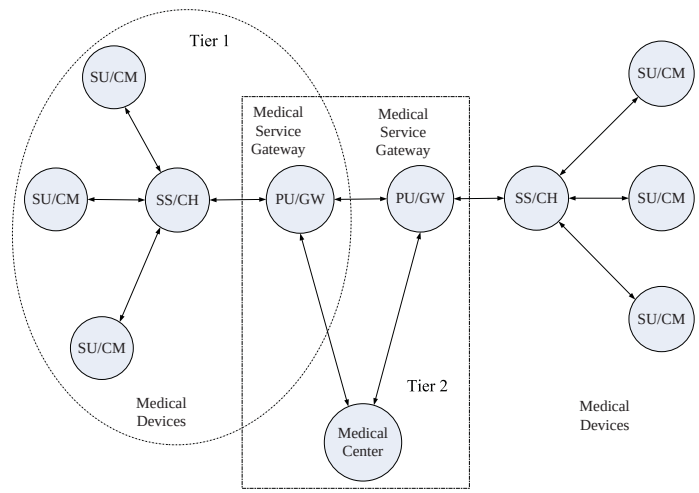

Fig. 3. A two-tier identification framework.

We assume: (1) each sensor node in a cluster has the same internal device delay characteristic and (2) a pair of sensors are equipped with clocks (oscillators) that are assumed to be asynchronous in time. Considering the existence of malicious users during the initialization phase, malicious nodes may establish their MIT tables via *Hello* messages exchanged with their neighboring nodes, which may help MUs achieve time synchronization in the network. However, based on the proposed scheme and simulation results, the MU can be filtered out using device information. Thus, the character of the sensors and the device delays are used to achieve node identification. Consequently, the proposed scheme can restrict the impact of the attacker on the basis of synchronization precision and device delay.

Note that these assumptions may be applied to healthcare scenarios or habitat monitoring applications such as locating patients or animals and performing intruder detection. Referring to a healthcare IoT scenario and Figure 2, a two-tier identification framework is depicted in Figure 3, including intra-cluster node identification and (2) inter-cluster node identification. The operations of T^2DAP protocol are described in the following subsections. Observe that on the basis of the task of each node, a clusterhead, a cluster member, and a gateway may play the role of a SS, a SU, and a PU, respectively. For depicting the proposed authentication system, Table I describes the system parameters.

A. Network Initialization

Before describing the forward and backward communication procedures between a PU and a SS, denote t_i^{ss} as the time stamp in sensor ss; denote d_{tx}^{ss} and d_{rx}^{ss} as the delay times of internal device for transmission and reception in sensor ss, respectively; let t_{del}^{ss} be the response delay in sensor ss; t_{ss_pu}

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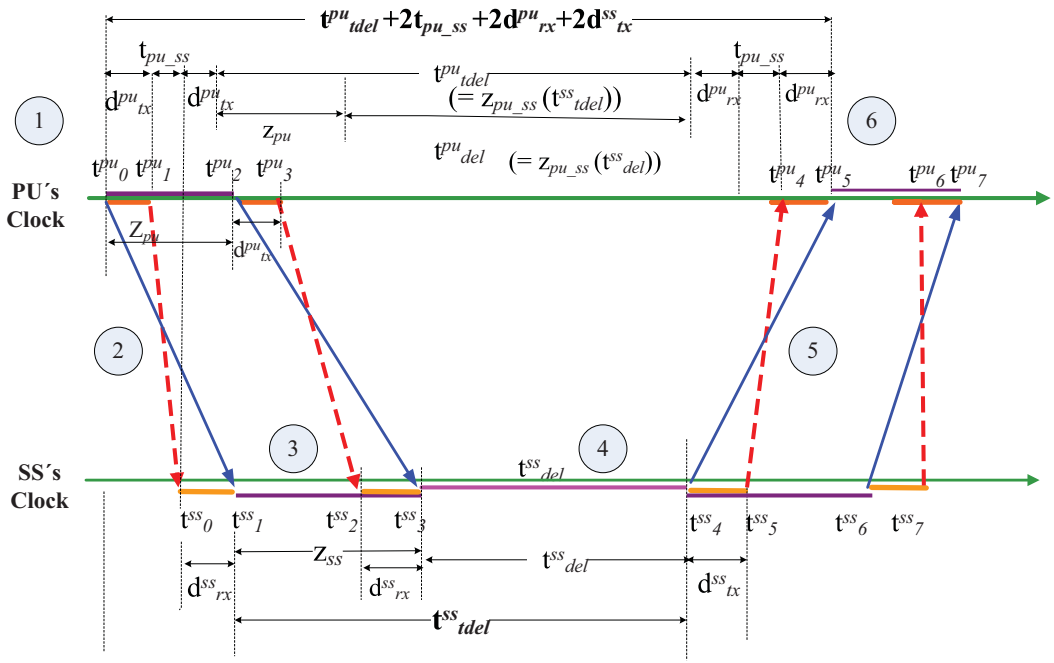

Fig. 4. The forward and backward communication between a PU and a SS of the T²DAP scheme.

is the signal propagation time between sensor ss and sensor pu. Note that the above notations apply to all nodes in the network. Figure 4 details the procedures of the forward and backward communication between a PU and a SS, which yield:

1) Sensor pu sends a message to sensor ss. Note that t_0^{pu} is the time of sending the first bit of the message from a sender, t_2^{pu} is the time of sending the last bit of the message from a sender. Thus, the message length is $z_{pu} = t_2^{pu} - t_0^{pu}$.
2) Due to the device delay, the actual message sending time stamps of sensor pu are $t_1^{pu} = t_0^{pu} + d_{tx}^{pu}$ and $t_3^{pu} = t_2^{pu} + d_{tx}^{pu}$.
3) Similarly, the message receiving time stamps of sensor ss are $t_1^{ss} = t_0^{ss} + d_{rx}^{ss}$ and $t_3^{ss} = t_2^{ss} + d_{rx}^{ss}$. Therefore, the time interval of receiving the message is $z_{ss} = t_3^{ss} - t_1^{ss}$.
4) After the response delay time t_{del}^{ss}, sensor ss transmits a message back to sensor pu along with the time stamp t_4^{ss} (the time on sensor ss's clock when it transmits).
5) The delay time of sensor pu is $t_{tdel}^{pu} = z_{pu_ss} \cdot (z_{ss} + t_{del}^{ss})$, where the scale factor $z_{pu_ss} = (t_2^{pu} - t_0^{pu})/(t_3^{ss} - t_1^{ss})$.
6) The propagation time t_{pu_ss} can be calculated as

$$t_{pu_ss} = \frac{[t_5^{pu} - t_0^{pu} - d_{\mathrm{de}}^{pu} - t_{tdel}^{pu} - z_{pu_ss} \cdot d_{\mathrm{de}}^{ss}]}{2} \quad (1)$$

$$\approx \frac{[t_5^{pu} - t_0^{pu} - 2d_{\mathrm{de}}^{pu} - t_{tdel}^{pu}]}{2}, \quad (2)$$

where the time stamp t_5^{pu} in sensor pu is based on the reception time of the message sent from sensor ss; $d_{\mathrm{de}}^{pu} = d_{tx}^{pu} + d_{rx}^{pu}$, $d_{\mathrm{de}}^{ss} = d_{tx}^{ss} + d_{rx}^{ss}$, and $d_{\mathrm{de}}^{pu} \approx z_{pu_ss} \cdot d_{\mathrm{de}}^{ss}$ due to the clock and device characteristics. Thus, the distance is $d_{pu_ss} = c \cdot t_{pu_ss}$ with the speed of the signal c.

After the sensors are deployed, each sensor broadcasts a *Hello* message for information exchange. A *Hello* message consists of: (1) RxID - the sensor ID of the receiving sensor, (2) TxID - the sensor ID of the sending sensor and (3) time stamps of the message transmission. Based on the *Hello* message, the measurement information table (MIT) is established, which consists of: (1) TxID - the sensor ID of the sending sensor, (2) RxID - the sensor ID of the receiving sensor, (3) SynZ - the scale factor of a clock due to time asynchronization between a pair of sensors, (4) RM - the length of received message, (5) PT - the estimated propagation time, and (6) DM - the distance measurement. Based on the handshaking procedure in Figure 4, Table II details the message format for information exchange between sensor pu and sensor ss. Notice that in Table II the values in Roman font represent the measurements via two-way communication (i.e. $pu \rightarrow ss \rightarrow pu$) and the values in bold font represent the measurements via three-way communication (i.e. $pu \rightarrow ss \rightarrow pu \rightarrow ss$). Table III illustrates the detailed MIT for the network initialization phase. The same operations can be applied to the communication between a SS and a SU or between a PU and a SU during the network initialization phase.

B. Intra-Cluster (Tier 1) Authentication

Figure 5 describes the communication mechanism of the T²DAP method for establishing the measurement information table and performing the detection for a PUEA attack. Assume

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TABLE I
DESCRIPTION OF SYSTEM PARAMETERS

Parameter	Description
t_i^{ss}	the time stamp in sensor ss at time step i
d_{tx}^{ss}/d_{rx}^{ss}	the delay time of internal device for transmission/reception in sensor ss
t_{ss_pu}	the signal propagation time between sensors ss and pu
t_{del}^{ss}	the response delay in sensor ss
t_{tdel}^{pu}	the delay time of sensor pu
z_{pu_ss}	the scale factor due to time asynchronization between sensor ss and sensor pu
z_{pu}	the message length of sensor pu
$d_{tx(\text{spec})}^{ss}$	the true device delay for transmission of sensor ss
$d_{rx(\text{spec})}^{ss}$	the true device delay for reception of sensor ss
$\hat{d_{\text{de}}^{ss}}$	the estimated device delay for transmission/reception of sensor ss
d_{spec}^{ss}	the sum of $d_{tx(\text{spec})}^{ss}$ and $d_{rx(\text{spec})}^{ss}$
n	the scale factor for the purpose of using a variable message length
η_{ss}	the metric calculated for the sensing decision of sensor ss
η_r	a given threshold trust value for node identification
α_{mu}	the scale factor for the device delay of sensor mu referenced to sensor ss

TABLE II
MEASUREMENT INFORMATION TABLE: PU ↔ SS

Sensor pu	TxID	RxID	SynZ	RM	PT	DM
	ID_{pu}	ID_{ss}	z_{ss_pu}	z_{pu}	$\mathbf{t_{pu_ss}}$	$\mathbf{d_{pu_ss}}$
Sensor ss	**TxID**	**RxID**	**SynZ**	**RM**	**PT**	**DM**
	ID_{ss}	ID_{pu}	$\mathbf{z_{pu_ss}}$	$\mathbf{z_{ss}}$	t_{ss_pu}	d_{ss_pu}

TABLE III
MEASUREMENT INFORMATION TABLE: NETWORK INITIALIZATION

Sensor pu	TxID	RxID	SynZ	RM	PT	DM
	ID_{pu}	ID_{ss}	z_{ss_pu}	z_{pu}	$\mathbf{t_{pu_ss}}$	$\mathbf{d_{pu_ss}}$
	ID_{pu}	ID_{su}	z_{su_pu}	z_{pu}	$\mathbf{t_{pu_su}}$	$\mathbf{d_{pu_su}}$
Sensor ss	**TxID**	**RxID**	**SynZ**	**RM**	**PT**	**DM**
	ID_{ss}	ID_{pu}	$\mathbf{z_{pu_ss}}$	$\mathbf{z_{ss}}$	t_{ss_pu}	d_{ss_pu}
	ID_{ss}	ID_{su}	$\mathbf{z_{su_ss}}$	$\mathbf{z_{ss}}$	t_{ss_su}	d_{ss_su}
Sensor su	**TxID**	**RxID**	**SynZ**	**RM**	**PT**	**DM**
	ID_{su}	ID_{pu}	$\mathbf{z_{pu_su}}$	$\mathbf{z_{su}}$	t_{su_pu}	d_{su_pu}
	ID_{su}	ID_{ss}	$\mathbf{z_{ss_su}}$	$\mathbf{z_{su}}$	t_{su_ss}	d_{su_ss}

the spectrum decision of PU is inactive and a MU claims to be a PU. Referring to Figure 5 (middle) (i.e. with a SS only), the procedures of node identification are depicted as follows:

1) Sensor ss transmits a sequence with length $z_{ss}^{'} = t_2^{ss'} - t_0^{ss}$, which is *n* times of the sequence length of sensor ss in Network initialization. At the same time, sensor

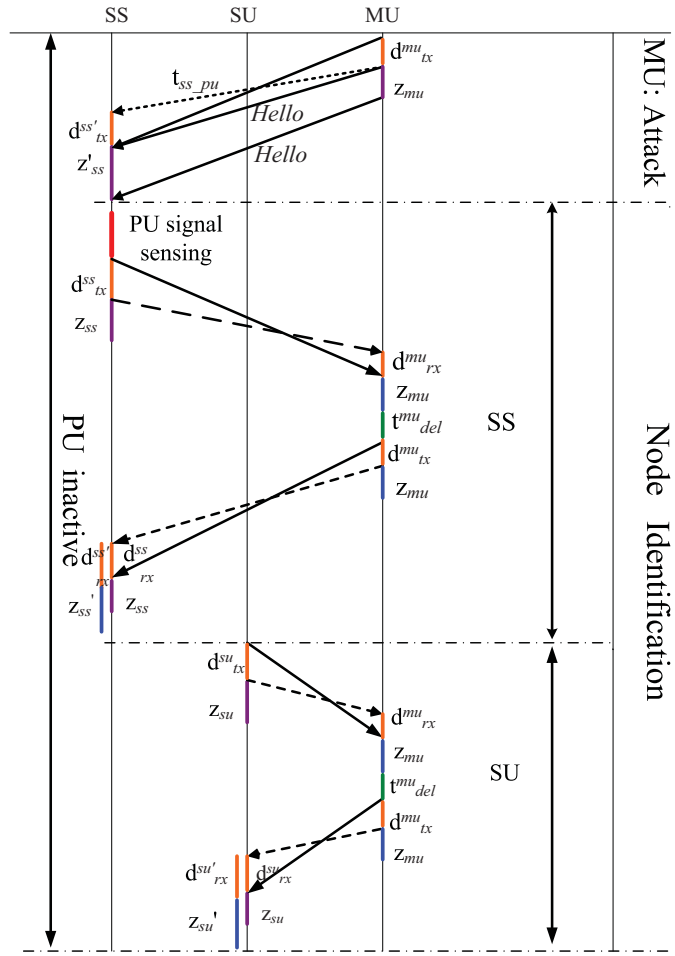

Fig. 5. The operating procedures defending against the PUEA attack in a cluster.

ss sends a message $(\text{ID}_{pu}, \text{ID}_{ss}, t_0^{ss}, t_2^{ss})$ to sensor mu with a wrong message sending time stamp of sensor ss, t_2^{ss}, where $z_{ss} = t_2^{ss} - t_0^{ss}$. That is, $z_{ss}^{'} = n \cdot z_{ss}$ with a real number *n*, which depends on sensor ss's clock. Note that the purpose of the variable message length is to confound the masquerade attack; the term is used for the purpose of authentication and is not related to information-theoretic criteria.

2) Assume the time interval of receiving the message of sensor mu is $z_{mu}^{'}$. The delay t_{tdel}^{mu} is given by

$$t_{tdel}^{mu} = z_{mu}^{'} + t_{del}^{mu}, \tag{3}$$

3) Accordingly, sensor mu calculates $z_{ss_mu}^{(n)} = z_{ss}/z_{mu}^{'}$. Note that z_{ss_mu} should be $z_{ss}^{'}/z_{mu}^{'}$ (i.e. $z_{ss_mu}^{(n)} = z_{ss_mu}/n$). Afterwards, sensor mu replies the message $(\text{ID}_B, \text{ID}_A, t_4^{mu}, t_{del}^{mu})$ to sensor ss at the time t_4^{mu} with

$$t_{tdel}^{ss} = z_{ss_mu}^{(n)} \cdot t_{tdel}^{mu} = \frac{z_{ss}}{z_{mu}^{'}} \cdot (z_{mu}^{'} + t_{del}^{mu}) \tag{4}$$

$$= \frac{z_{ss}^{'}}{n}(1 + \frac{t_{del}^{mu}}{z_{mu}^{'}}). \tag{5}$$

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4) Thus, referring to (2), the estimated device delay for transmission and reception of sensor ss yields

$$\hat{d_{\text{de}}^{ss}} = t_5^{ss} - t_0^{ss} - t_{tdel}^{ss} - 2 \cdot t_{ss_pu} - d_{\text{spec}}^{ss}, \quad (6)$$

where $d_{\text{spec}}^{ss} = d_{tx(\text{spec})}^{ss} + d_{rx(\text{spec})}^{ss}$ and $d_{tx(\text{spec})}^{ss}$ and $d_{rx(\text{spec})}^{ss}$ are the true device delays for transmission and reception of sensor ss, respectively. Note that the time stamp t_5^{ss} is based on the reception time of the message sent from sensor mu and t_{ss_pu} is known from sensor mu's MIT table.

5) Decision criterion:
   - If $\eta_{ss} \leq \eta_r$, then sensor mu passes the node identification test.
   - Otherwise, sensor mu is regarded as an anti-node.

   η_{ss} is defined as:

$$\eta_{ss} = |(\hat{d_{\text{de}}^{ss}} - d_{\text{spec}}^{ss})/d_{\text{spec}}^{ss}|, \quad (7)$$

   where η_r is a given threshold trust value for node identification. In the general case, $\eta_r \leq 1$.

Note that the decision criterion is based on the estimated device delay and the device information of PUs (Gateways) and SSs (sensing devices), which is related to the measurement noise, minimum detectable signal power, and timing resolution. As detailed in Section V, the detection quality is related to timing resolution of the clock, time synchronization, the threshold value, and device delay of a node, which suggests that given clock information and device information, an appropriate value of the detection threshold should be selected in order to perform the identification mechanism.

Instead of using a SS to detect a PUEA attack, one possible cooperative scheme is to perform centralized decision fusion. After the sensing sensor forwards the spectrum sensing decision, the SUs may response to different sensing decisions of the MU. Afterward, the sensing sensor may request the SUs to report their MU decisions, and then use a majority vote to decide whether the MU is an anti-node or not. Thus, a node group (a SS and a set of SUs) may work together and follow Steps 1 $\sim$ 5 depicted above to verify a MU in a cooperative manner. Assume M is the number of SUs to perform cooperative identification. The decision criterion with majority voting for the MU to pass the node identification test is

$$A_{ss} + \sum_{i=1}^{M} B_{su}^{(i)} > \frac{M+1}{2}, \quad (8)$$

where A_{ss} and $B_{su}^{(i)}$ are indicator random variables, which are

$$A_{ss} = \mathbf{1}_{\{\eta_{ss} \leq \eta_r\}} \quad (9)$$

$$B_{su}^{(i)} = \mathbf{1}_{\{\eta_{su}^{(i)} \leq \eta_r\}}. \quad (10)$$

Notice that the metric is calculated by the sensing decisions of the SS and SUs. The sensing sensor will collect the sensing decisions of SUs and then calculate this metric to make a detection decision. Similarly, η_{su} is given by $\eta_{su} = |(\hat{d_{\text{de}}^{su}} - d_{\text{spec}}^{su})/d_{\text{spec}}^{su}|$. Thus, the MU nodes may be cooperatively verified by the node group, a SS and SUs. Figure 6 depicts the virtual flow chart of intra-cluster (tier 1) authentication.

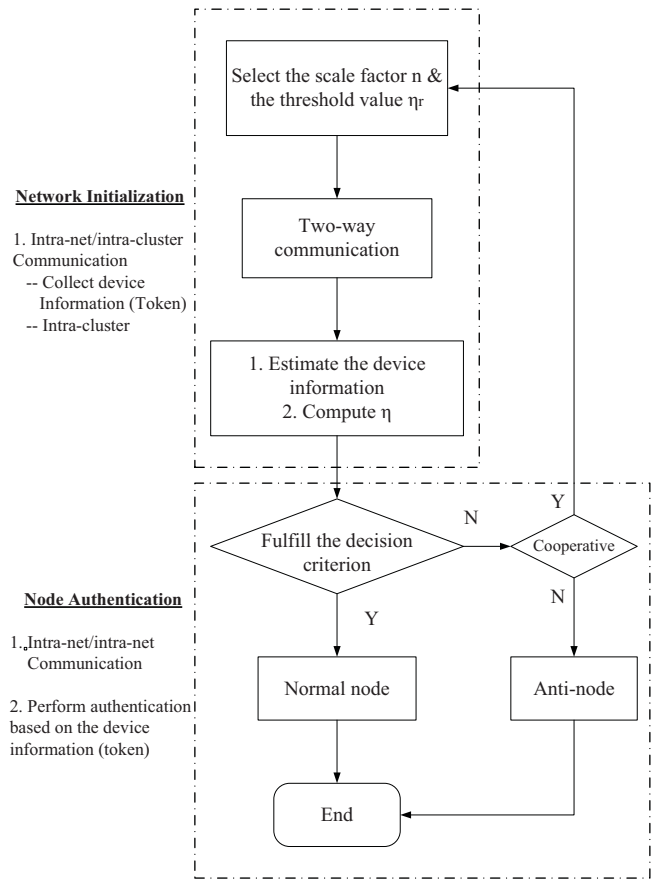

Fig. 6. The virtual flow chart of intra-cluster (tier 1) authentication.

C. Inter-Cluster (Tier 2) Authentication

In Tier 1, an authenticated mechanism is applied to identify the anti-nodes in a cluster. Two delay characteristics (e.g., a SU and a PU) can be distributed locally under cluster construction for authentication, which implies that these two device delays may be analogous to symmetric keys (e.g., a cluster-member key and a gateway key) such that the securities of intra- and inter-cluster communication are established upon the device information. Therefore, similar to the process of key distribution shown in Figure 7 [30], we may consider the SU delay information and the PU delay information as a cluster-member key and a gateway key, respectively.

In Tier 2, considering a PU-PU communication scenario, PU_1 (e.g., a medical service gateway) initiates a communication process and sends its ID and receiver's ID (e.g., PU_2). Next, the receiver PU_2 replies a message to PU_1. As detecting the PU's activity, the SS and SUs are forced to be inactive. Afterwards, PU_1 applies the device information to compute the secure token, which may be defined by (4), considering the response delay time of PU_2, $t_{del}^{PU_2}$. Since $t_{del}^{PU_2}$ is critical for calculating the estimated device delay, this parameter may be manipulated to confuse the anti-nodes. For instance, we may use $t_{del}^{PU_2}$ and k, which are obtained in the network initialization phase, as a basis to set $t_{del}^{PU_1} = k \cdot t_{del}^{PU_2}$, where

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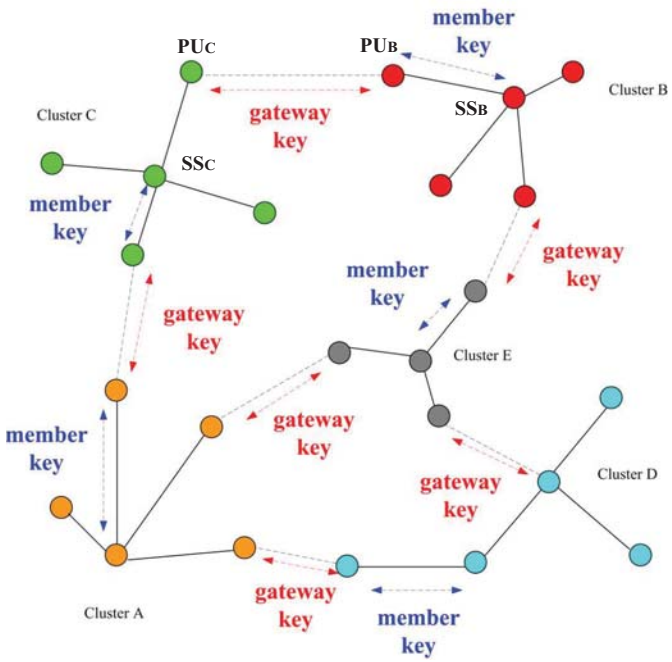

Fig. 7. The conceptual network structure of device information (key) distribution for IoT applications.

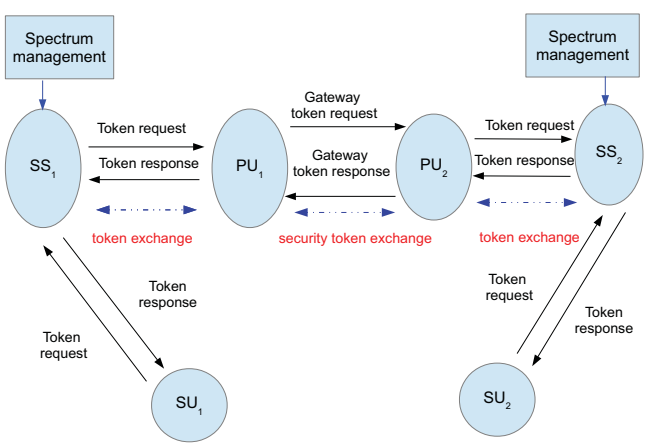

Fig. 8. The conceptual information flow of $\mathrm{T}^2\mathrm{DAP}$.

$k > 0$. Accordingly, a specific $t_{del}^{PU_2}$ can be applied to generate a corresponding $t_{del}^{PU_1}$ for the next communication process. Therefore, if the token is valid, then PU_2 is determined to be a trustable gateway for further information exchange. Notice that the above operations do not involve any encryption/decryption.

Following the operation procedures of $\mathrm{T}^2\mathrm{DAP}$ scheme, Figure 8 describes the conceptual information flow of the proposed mechanism. Observe that the token information exchange can be performed from three scenarios, identified as a PU-PU communication, a PU-SS communication, and a SS-SU communication.

IV. Performance Analysis

A. Security Analysis

This section performs a security analysis to examine the proposed protocol, which uses three important features (clock, distance, and device characteristics) to establish the countermeasure. Four scenarios of a MU with respect to the PU with separate sensing are used to assess the capability of $\mathrm{T}^2\mathrm{DAP}$: (1) Case I - without synchronous clocking and without device characteristics; (2) Case II - without synchronous clocking and with device characteristics; (3) Case III - with synchronous clocking and without device characteristics; (4) Case IV - with synchronous clocking and with device characteristics. Without loss of generality, we may assume that the order of timing resolution is nanosecond (ns), the order of device delay is about $10 \sim 100$ ns [31], and the distance between the PU and a SS is from 1 m to 100 m, which implies that the order of propagation time is between 3.3 ns and 333.3 ns. Referring to detection criterion (7) and considering the above settings, a specific range of η_r may be selected for verifying the malicious nodes. For instance, as the order of device delay is close to that of propagation time, we may select η_r of small enough magnitude to satisfy the detection requirement; otherwise, the larger values of η_r may be adopted. Consequently, the device delay and distance between the PU and a MU may have a significant impact on the node identification. Section V compares the proposed method with the energy detection (ED) scheme [11], which derives a lower bound on the probability of a successful PUEA. Moreover, a BP scheme presented in [10] is applied to derive a benchmark for the proposed scheme. The performance evaluation shows that the detection capability of proposed scheme is superior to that of these methods.

B. Decision Criterion (η) Distribution Analysis

Considering the device delay and distance measurement, the decision criterion may be analyzed as a function of the clock accuracy by deriving an approximate distribution for the estimation.

1) Clock-Dependent Ranging and Device-Delay Measurement Accuracy: The random variable T represents the sensor estimate of the true time *t*; thus T_{ss_pu} is an estimate of the true time t_{ss_pu} and T_i^{ss} is the estimate of the time t_i^{ss} as measured by the clock of sensor ss. Assume that sensor *ss* and sensor *pu* have the same device information or sender *pu*'s characteristics (e.g., device delay) is known to sensor *ss*. Let D_{tx}^{ss}, D_{rx}^{ss} and D_{de}^{ss} be independent normal random variables, where $D_{de}^{ss} = D_{tx}^{ss} + D_{rx}^{ss}$. Referring to (2), the estimated device delay time for transmission of sensor *ss* is

$$D_{tx}^{ss} = \frac{[T_5^{ss} - T_0^{ss} - T_{tdel}^{ss} - 2T_{ss_pu}]}{4}. \tag{11}$$

$$T_{tdel}^{ss} = Z \cdot T_{tdel}^{pu}, \tag{12}$$

$$T_{del}^{pu} = T_4^{pu} - T_1^{pu}, \tag{13}$$

where

$$Z = \frac{T_2^{ss} - T_0^{ss}}{T_3^{pu} - T_1^{pu}} \tag{14}$$

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is a scale factor that represents how much faster or slower clock *ss* moves than clock *pu*.

For the purpose of analysis, assume that all measurements T_i^{ss} and T_j^{pu} are independent normal random variables with the same variance σ_r^2 caused by the measurement error in the clock:

$$T_i^{ss} \sim N(t_i^{ss}, \sigma_r^2) \quad \text{for } i = 0, 1, 2, \ldots, 7. \tag{15}$$

$$T_j^{pu} \sim N(t_j^{pu}, \sigma_r^2) \quad \text{for } j = 0, 1, 2, \ldots, 7. \tag{16}$$

This normality assumption is justified in [32] when the clock skew is small. Hence the random variable Z is the ratio of two normal random variables. As shown in [33] and [34], under reasonable conditions on the distributions, Z is well approximated by

$$Z \sim N(\mu_Z, \sigma_Z^2) \tag{17}$$

with

$$\mu_Z = \frac{\mu_{z_{ss}}}{\mu_{z_{pu}}}$$

and

$$\sigma_Z^2 = \frac{2\sigma_r^2}{\mu_{z_{pu}}^2}\left(1 + \left(\frac{\mu_{z_{ss}}}{\mu_{z_{pu}}}\right)^2\right),$$

where $\mu_{z_{ss}} = t_2^{ss} - t_0^{ss}$ and $\mu_{z_{pu}} = t_3^{pu} - t_1^{pu}$. For this Gaussian approximation to hold, $\mu_{z_{pu}}$ must be biased away from zero and the ratio μ_{z_b}/σ_r^2 must be large. These are reasonable assumptions in the sensor communication application.

From (12) and (17), T_{tdel}^{ss} can be viewed as the product of two normal random variables. Since the measurement errors are small, [35] shows that the distribution of T_{tdel}^{ss} can be sensibly approximated by

$$T_{tdel}^{ss} \sim N\left(\mu_Z t_{tdel}^{pu}, 2\mu_Z^2\sigma_r^2 + {t_{tdel}^{pu}}^2\sigma_Z^2\right) \tag{18}$$

when μ_Z/σ_Z and $\mu_{T_{tdel}^{pu}}/\sigma_{T_{tdel}^{pu}}$ are large, which are reasonable assumptions in this case. Based on (1), we have

$$D_{ss_pu} \sim N\left(\mu_{D_{ss_pu}}, \sigma_{D_{ss_pu}}^2\right), \tag{19}$$

$$T_{ss_pu} \sim N\left(\mu_{T_{ss_pu}}, \sigma_{T_{ss_pu}}^2\right), \tag{20}$$

where $\mu_{T_{ss_pu}} = \mu_{D_{ss_pu}}/c$ and $\sigma_{T_{ss_pu}}^2 = \sigma_{D_{ss_pu}}^2/c^2$. To simplify the analysis, let D_{tx}^{ss} and D_{rx}^{ss} be identical independent normal random variables. Using the above analysis and referring to (11), the distribution of D_{de}^{ss} is

$$D_{de}^{ss} \sim N\left(\mu_{D_{de}^{ss}}, \sigma_{D_{de}^{ss}}^2\right), \tag{21}$$

where

$$\mu_{D_{de}^{ss}} = \frac{1}{2}(t_5^{ss} - \mu_Z t_{tdel}^{pu} - t_0^{ss} - 2t_{ss_pu})$$

and

$$\sigma_{D_{de}^{ss}}^2 = \frac{1}{4}\left[(2 + 2\mu_Z^2)\sigma_r^2 + {t_{tdel}^{pu}}^2\sigma_Z^2 + 4(\sigma_{T_{ss_pu}}^2)\right].$$

Note that the mean of random variable D_{de}^{ss} is the true value of the device delay time and the variance of D_{de}^{ss} depends on the variance of the propagation time, the timing resolution σ_r^2, the characteristic of clock-adjustment factor (14), and the time delay t_{tdel}^{pu}.

Let a random variable $X = D_{de}^{ss}/d_{\text{spec}}^{ss} - 1$. Thus, we have $\eta_{ss} = |X|$. Referring to (7) and (21), the distribution of X is

$$X \sim N\left(\mu_X, \sigma_X^2\right). \tag{22}$$

where

$$\mu_X = \frac{\mu_{D_{de}^{ss}}}{d_{\text{spec}}^{ss}} - 1$$

and

$$\sigma_X^2 = (d_{\text{spec}}^{ss})^{-2}\sigma_{D_{de}^{ss}}^2.$$

Accordingly, the distribution of the decision criterion η_{ss} is given by

$$f_{\eta_{ss}}(x; \mu_X, \sigma_X^2) = \sqrt{\frac{2}{\pi\sigma_X^2}}\exp^{-\frac{(x^2+\mu_X^2)}{2\sigma_X^2}}\cosh\left(\frac{\mu_X x}{\sigma_X^2}\right). \tag{23}$$

with

$$\mu_{\eta_{ss}} = \sigma_X\sqrt{\frac{2}{\pi}}\exp\left(\frac{-\mu_X^2}{2\sigma_X^2}\right) - \mu_X \text{erf}\left(\frac{-\mu_X}{\sqrt{2\sigma_X^2}}\right)$$

$$\sigma_{\eta_{ss}}^2 = \mu_X^2 + \sigma_X^2 - \mu_{\eta_{ss}}^2$$

for $x \geq 0$, and 0 everywhere else, where $\text{erf}(\cdot)$ is the error function and $\cosh(\cdot)$ is cosine Hyperbolic function. It follows that the cumulative distribution function is given by

$$F_{\eta_{ss}}(x; \mu_X, \sigma_X^2) = \frac{1}{2}\left[\text{erf}\left(\frac{x + \mu_X}{\sqrt{2\sigma_X^2}}\right) + \text{erf}\left(\frac{x - \mu_X}{\sqrt{2\sigma_X^2}}\right)\right] \tag{24}$$

for $x \geq 0$. Note that both μ_X and σ_X^2 can be applied to depict the characteristics of η_{ss} in the folded distribution.

Figure 9 illustrates the folded distribution of the decision criterion. Assume that the device delay of an anti-node MU, d_{spec}^{mu}, is referenced to d_{spec}^{ss}, which yields $d_{\text{spec}}^{mu} = \alpha_{mu} \cdot d_{\text{spec}}^{ss}$, where $\alpha_{mu} > 0$. Note that with $\alpha_{mu} = 1$ (i.e., with the same device information), we have $\mu_X = 0$, which leads the folded distribution of the decision criterion to a half-normal distribution. Observe that given $\eta_r = 0.1$ and different device characteristics of a MU (i.e., with different values of α_{mu}), the area under the curve of folded distribution for the interval $x \geq \eta_r$ represents the successful attack detection rate, which implies that compared with the attack scenario of $\alpha_{mu} = 1$, the attack scenario of $\alpha_{mu} = 0.5$ yields a higher successful detection rate. Section V-A3 contains detailed discussions on detection performance.

C. Energy Detection: Lower Bound of A Successful PUEA

This subsection is presented to assess the detection performance comparison. Regarding the detection of the PUEAs, received power measurements of the SUs can be applied to detect the attack. On the basis of the results in [11], a lower bound on the probability of a successful PUEA on a SS by a set of co-operating MUs is derived.

Suppose one MU masquerades a PU. Since the received power of the ith SS is different from the PU transmitter and a MU, it could be used to identify the power of the transmitted signal whether from the PU transmitter or a MU. A two-ray ground model for the signal from the PU or the MU is

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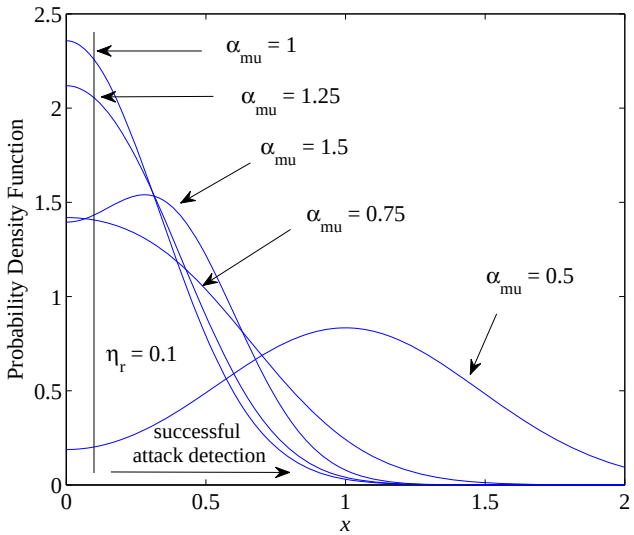

Fig. 9. The folded distribution of the decision criterion, considering device characteristics between a SS and a MU.

considered. Therefore, due to the effects of multipath, both the path loss exponent for the propagation from the PU to any SU and that between any MU and any SU are four. Define $P_r^{(pu)}(i)$

$$P_r^{(pu)}(i) = P_t (d_{pu}^{(i)})^{-4} (G_{pu}^{(i)})^2 (R_{pu}^{(i)})^2, \tag{25}$$

where $d_{pu}^{(i)}$ is the distance from the PU transmitter to the ith SS, $(G_{pu}^{(i)})^2$ is the log-normal shadowing from the PU transmitter to the ith SS and $R_{pu}^{(i)}$ is the Rayleigh fading from the PU transmitter to the ith SS. Referring to the assumptions in [11], the Rayleigh fading random variables from the primary transmitter and all malicious users to all secondary users are distributed with mean δ_{pu} and δ_{mu}, respectively. The received power at the ith SS from the MU transmitter, $P_r^{(mu)}(i)$, is given by

$$P_r^{(mu)}(i) = P_t (d_{mu}^{(i)})^{-4} (G_{mu}^{(i)})^2 (R_{mu}^{(i)})^2. \tag{26}$$

Combining (25) and (26), the detection rule is

$$|P_r^{(pu)}(i) - P_r^{(mu)}(i))| \leq \epsilon, \tag{27}$$

where ϵ is a constant threshold value for node identification. Thus, the probability of a successful PUEA is

$$P_{PUEA} = Pr\{|P_r^{(pu)}(i) - P_r^{(mu)}(i))| \leq \epsilon\}. \tag{28}$$

Note that in order to simplify the analysis, the Markov inequality is applied, which implies that the probability P_{PUEA} in (28) can be bounded as

$$P_{PUEA} \geq 1 - \frac{|E[P_r^{(pu)}(i)] - E[P_r^{(mu)}(i)]|}{\epsilon}. \tag{29}$$

To evaluate the expectations, Fenton's approximation is adopted. Accordingly, the mean $E[P_r^{(pu)}(i)]$ is given by

$$E[P_r^{(pu)}(i)] = P_t \Delta_{pu} e^{1/2} a^2 \sigma_{pu}^2 E[(d_{pu}^{(i)})^{-2}], \tag{30}$$

where a $=(\ln 10)/10$, the distance $d_{pu}^{(i)}$ is given by

$$d_{pu}^{(i)} = \sqrt{r_i^2 + r_{pu}^2 - 2 r_i r_{pu} cos(\theta_i - \theta_{pu})}, \tag{31}$$

(r_i, θ_i) are the co-ordinates of the ith SS, and (r_{pu}, θ_{pu}) are the co-ordinates of the PU. Therefore, the $E[(d_{pu}^{(i)})^{-2}]$ can be further simplified as

$$E[(d_{pu}^{(i)})^{-2}] = \frac{1}{\pi R^2} \int_0^R \int_0^{2\pi} \frac{r_i dr_i d\theta_i}{r_i^2 + r_{pu}^2 - 2 r_i r_{pu} \cos(\theta_i - \theta_{pu})}. \tag{32}$$

By fixing the coordinate of a SU in $(0, 0)$ and substituting (32) into (30), the latter can be simplified by removing the integration, which yields

$$E[P_r^{(pu)}(i)] = \frac{P_t e^{\frac{1}{2}} a^2 \sigma_{pu}^2}{r_{pu}^2} \Delta_{pu}. \tag{33}$$

Similarly, referring to assumptions above and performance approximations made in [36], [37], $E[P_r^{(mu)}(i)]$ is given by

$$E[P_r^{(mu)}(i)] = \frac{1}{\pi(R^2 - R_0^2)} e^{\frac{1}{2}} a^2 \sigma_{mu}^2 \int_{R_i}^R \int_0^{2\pi} e^{a\hat{\mu}} r_j dr_j d\theta_j \tag{34}$$

$$= \frac{1}{2R_0^2(R^2 - R_0^2)} e^{\frac{1}{2}} a^2 \sigma_{mu}^2 \Delta_{mu}, \tag{35}$$

where the positions of SUs and MUs are distributed in the circular grid of radius R; R_0 is defined as the exclusive distance from the SU centered at (r, θ), which implies that for the SU fixed at coordinate (r, θ), no MUs are present within a circle of radius R_0. This is because that without this restriction, the received power at the SU due to transmission from any subset of MUs present within a distance R_0 from the SU becomes too large to create a PUEA [11]. Therefore, applying (33) and (35), we obtain the lower bound on a successful PUEA on a MU.

V. Simulation Results

The T^{2}DAP scheme is evaluated via simulations and numerical examples, considering measurement errors from several sources such as timing resolution, response delay, device delay, and clock calibration. The system performance under masquerade attack is investigated. Suppose that N_s normal sensor nodes in the two-dimensional space are deployed according to the uniform distribution in a square with the side length $l = 100$ meters. For the anti-nodes, assume that network instances comprise $N_A = 20$, 30, 40, 50, 60, 70, 80, 90 and 100 anti-nodes, respectively, which are deployed with a uniform distribution. Assume the device delay of an anti-node MU, d_{spec}^{mu}, is referenced to d_{spec}^{ss}, which yields $d_{\text{spec}}^{mu} = \alpha_{mu} \cdot d_{\text{spec}}^{ss}$, where $\alpha_{mu} > 0$. Without loss of generality, we investigate the typical performance given $\alpha_{mu} = 1$, the response delay $t_{del}^{pu} = 10^{-6}$ seconds, $z_{ss} = 1.83$ μs, $z_{pu} = 1.92$ μs (i.e. $z_{ss_pu} = 0.953$), $N_s = 30$, and $n = 1$. In this work, two kinds of signals, ultrasonic (US) signal and radio frequency (RF) signal, are examined for short-range communication and long-range communication applications, respectively. Referring to the node deployment, we explore the impacts of device delay, time synchronization, distance

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d_{pu_mu}, and threshold value η_r on system performance. The last set of experiments presents the performance comparison among the proposed T^2DAP, the belief propagation-based scheme [10] and energy detection-based scheme [11].

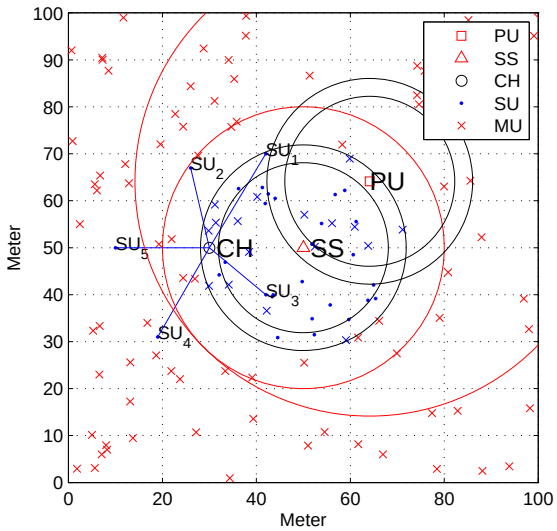

Fig. 10. An attack scenario for sensor ss and sensor pu.

A. Long-Range Communication Applications

Figure 10 demonstrates an attack scenario for possible long-range communication environments, assuming the transmission range is about 30 m. The study in [31] suggests that $4 \sim 45$ ns may be a reasonable value for the device delay measurement of a RF signal. Let the time unit of the device delay be one CPU clock time (e.g. CPU frequency: 2.4 GHz; 1 CPU clock time = 0.416 ns). In this work, the device delay is assumed to be 100 clock times (i.e., 41.6 ns).

1) The Impact of Device Delay and Time Synchronization: Figure 11 (top) shows the successful detection rate for varying the value of device delay and the number of malicious nodes. Referring to (7), given a threshold trust value $\eta_r = 0.1$, a smaller value of d_{spec} allows a smaller deviation range $|\hat{d}_{\mathrm{de}} - d_{\mathrm{spec}}|$, which increases the successful detection rate. Observe that due to the random uniform deployment of the anti-nodes, the successful detection rate maintains a small variation even with an increase in density of malicious nodes. Considering an anti-node with synchronous clocking and the same device characteristic ($\alpha_{mu} = 1$), the performance of the proposed secure scheme ($\approx 81\%$) is comparable to that of the belief propagation-based (BP) system in [10] ($\approx 80\%$). In contrast, considering an anti-node with different device characteristic ($\alpha_{mu} \neq 1$), the performance of the proposed secure scheme ($\approx 90\%$ for $\alpha_{mu} = 1.2$) is superior to that of the BP detection system. With the same settings as in the top, Figure 11 (middle) shows the successful detection rate when varying the time asynchronization of an anti-node. Observe that with varying the value of N_A, the device characteristic ($\alpha_{mu} = 1$ and $z_{mu_pu} = 0.995$) and ($\alpha_{mu} = 1$ and $z_{mu_pu} = 1.01$) lead to the detection rate of the proposed

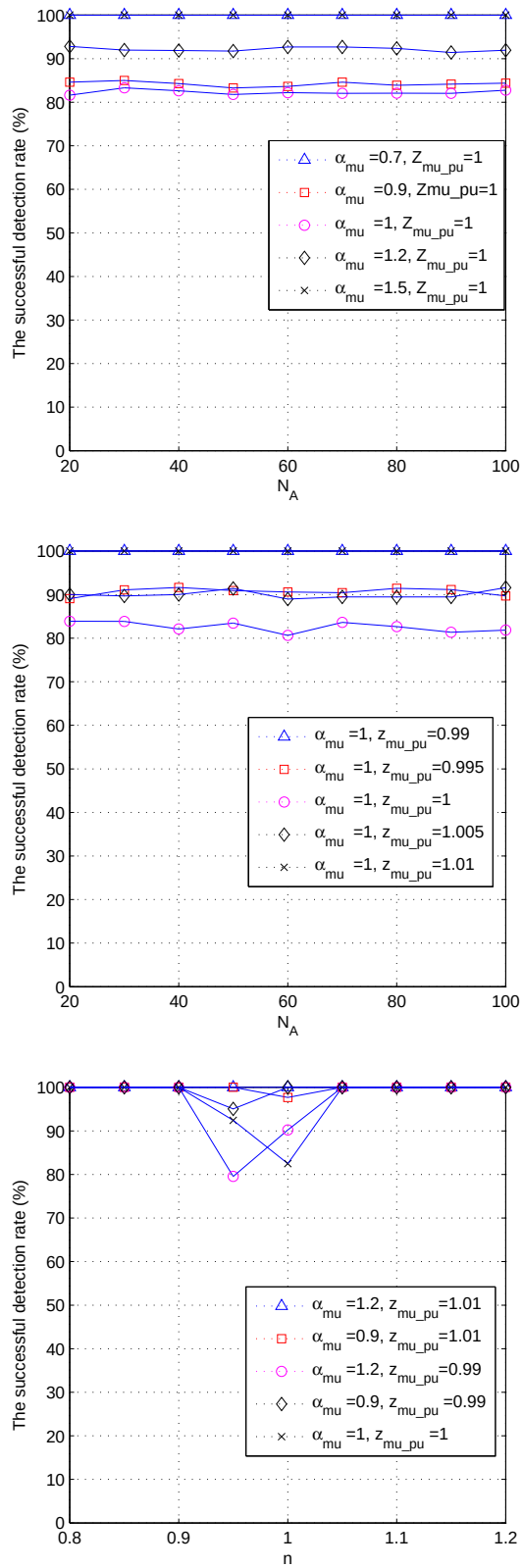

Fig. 11. The successful detection rate: with varying N_A and given the device delay of anti-nodes (top); with varying N_A and given the scale factor of anti-nodes (middle); with varying n and given the device and clock information (bottom).

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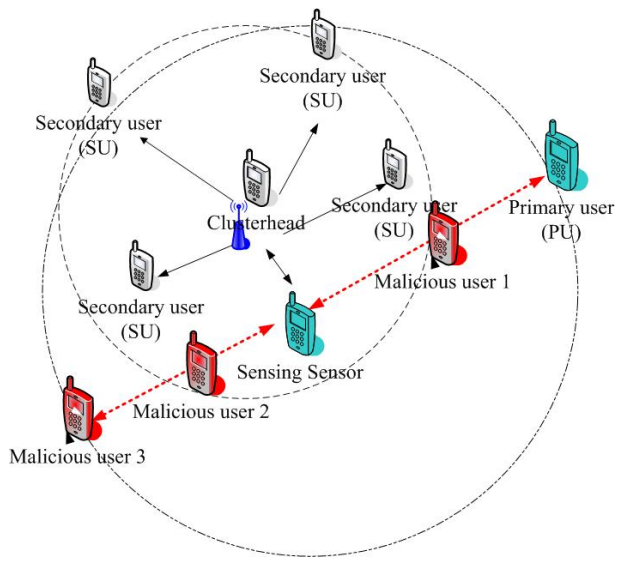

Fig. 12. An attack scenario for exploring the impact of the distance between a PU and a MU on system performance.

secure scheme $\approx 90\%$ and $\approx 100\%$, respectively. Therefore, comparing to that of an anti-node with time synchronization (e.g., $\approx 81\%$ for $\alpha_{mu} = 1$ and $z_{mu_pu} = 1$). Thus, time synchronization has an impact on detection performance and clock calibration could be used as a means of security method.

As depicted in Figure 11 (middle), given the device information, the successful detection rate of an anti-node without time synchronization (e.g., $\approx 90\%$ for $\alpha_{mu} = 1$ and $z_{mu_pu} = 0.995$) is higher than that of an anti-node with time synchronization (e.g., $\approx 81\%$ for $\alpha_{mu} = 1$ and $z_{mu_pu} = 1$). Thus, time synchronization has an impact on detection performance and clock calibration could be used as a means of security method.

Figure 11 (bottom) considers the successful detection rate with varying the amplification factor *n*. Referring to Case I (without synchronous clocking and without device characteristics) in Section IV-A, observe that the detection rate is a function of clocking, device information, and the scale factor. Moreover, due to the diversity of device characteristics and time synchronization, the detection performance of DBSS is concave upward from $n = 0.9$ to $n = 1.1$. This means the detection rate is bounded in a range (80%, 100%), which coincides with the results in the top and middle of Figure 11.

2) The Impact of the Distance d_{pu_mu}: Figure 12 shows an example of node deployment with normal nodes and anti-nodes, which depicts an attack scenario. Considering the actions of the malicious nodes (e.g. masquerading themselves as a PU), the protocol characteristics are explored. Given the threshold trust value ($\eta_r = 0.1$) and with synchronous clocking ($z_{mu_pu} = 1$), Figure 13 (top) explores the impact of device characteristic on system performance. As shown in Figure 13 (top), given $\alpha_{mu} = 1$, as the distance between the primary user and the PUEA attacker increases (e.g., $d_{pu_mu} \geq 1$ m), the successful detection rate is larger, which represents a higher probability of being a PUEA attacker. Observe that there exists

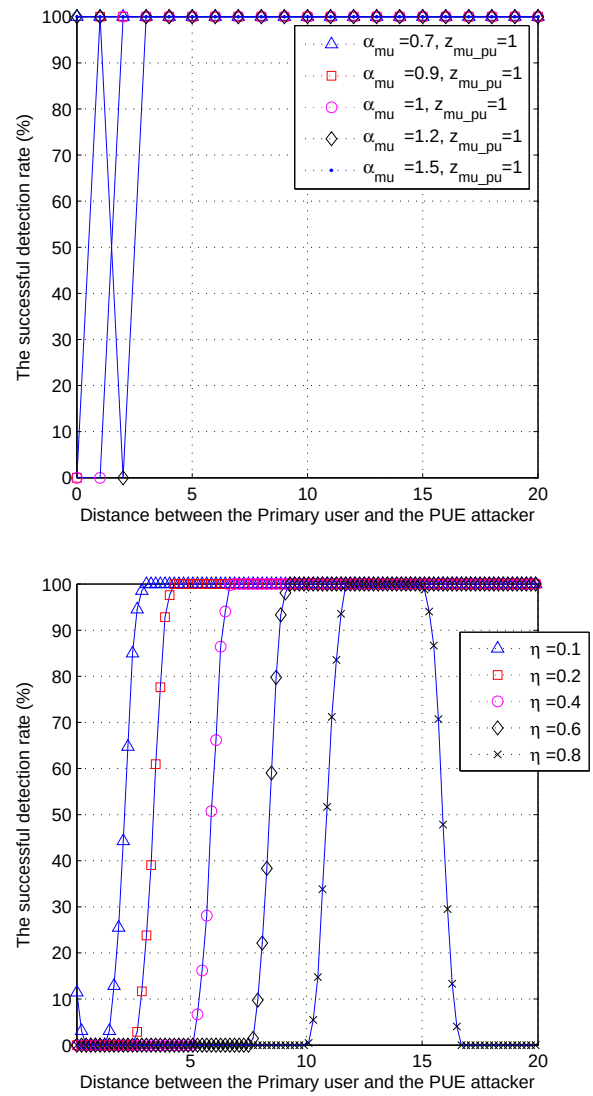

Fig. 13. The successful detection rate with varying the distance between a PU and a MU: given the device delay of anti-nodes (top) and given the threshold trust value (bottom).

a potential undetectable area (e.g., $d_{pu_mu} < 1$ m) due to the decision criterion in (7). Accordingly, with $\alpha_{mu} = 1$, we have an undetectable area about 1 m $> d_{pu_mu} > 0$ m. Similarly, given the device information ($\alpha_{mu} = 1$) and with synchronous clocking ($z_{mu_pu} = 1$), Figure 13 (bottom) explores the impact of the threshold trust value on node identification performance. As expected, a lower threshold trust value leads to a smaller undetectable area due to an increased strictness in the decision criterion (7). Notice that the blind detection range has drifted due to the device and clocking information of a MU with respect to the PU.

3) The Impact of Threshold Value η_r: Based on the settings in Figure 11, Figure 14 (top) considers the successful detection rate with varying the threshold η, the device delay of an anti-node α_{mu}, and the time asynchronization of an anti-node z_{mu_pu}. Observe that given $\eta_r = 0.1$ and with varying the value of α_{mu}, the successful detection rate of the proposed

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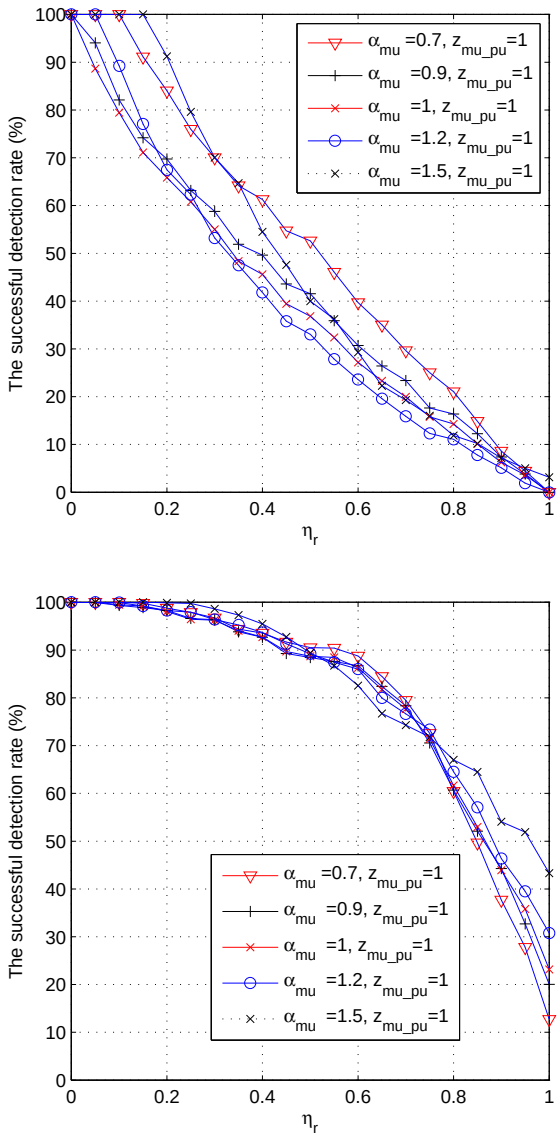

Fig. 14. The successful detection rate with varying η_r and the device delay of anti-nodes: non-cooperative method (top) and cooperative method with a SU (bottom).

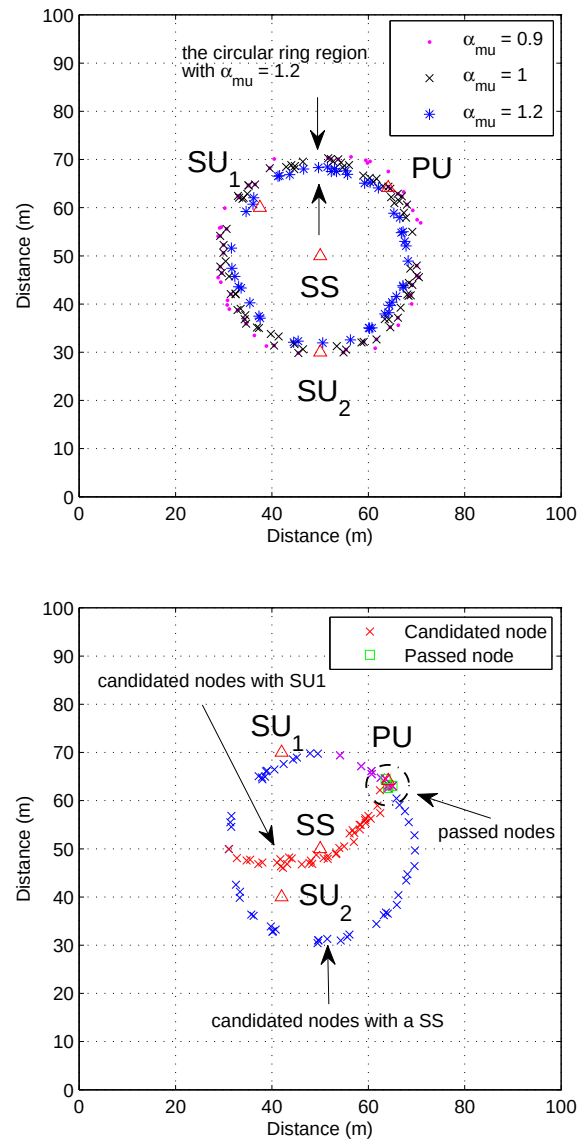

Fig. 15. The distribution of passed nodes with $\eta_r = 0.1$ and synchronous clocking: non-cooperative method (top) and cooperative method with a SU (bottom).

system achieves about 83% and 90% for the system with $\alpha_{mu} = 0.9$ and $\alpha_{mu} = 1.2$, respectively.

Figure 14 (bottom) shows the performance improvement with the cooperative identification scheme. Observe that given a device delay, a smaller threshold trust value of η_r leads to a higher successful detection rate. This is attributed to the fact that a smaller η_r contributes to a smaller passing range, which yields $(1-\eta_r)\cdot d_{\text{spec}} \leq \hat{d}_{\text{de}} \leq (1+\eta_r)\cdot d_{\text{spec}}$. That means changes in the value of η_r affect the passing area (i.e. the band width of a circular ring). Given the number of anti-nodes $N_A = 30$, $n = 1$, $\eta_r = 0.1$, and considering anti-nodes with the same device characteristic (i.e. $\alpha_{mu} = 1$) and synchronous clocking with respect to sensor SS, the successful detection rate improves from 83% with the non-cooperative scheme in Figure 14 (top) to 99% with the cooperative scheme associated applying the node group, a SS and a SU, in Figure 14 (bottom).

In order to visualize the detection performance and assess the effectiveness of the cooperative scheme, Figure 15 shows the distributions of passed nodes with non-cooperative and cooperative methods, respectively. Given $\eta_r = 0.1$, Figure 15 (top) depicts that the device characteristic has an impact on determining the inner radius of passing area of a circular ring. Thus, in order to pass the authentication test, a MU must be smart enough to locate itself within the corresponding passing area. Note that since there is no angle information of the measurement, the passing area of a SS (or a SU) is a circular ring. Compared with the non-cooperative scheme, Figure 15 (bottom) suggests that the cooperative scheme significantly suppresses the passing area against a PUEA attack. Observe that the passed nodes are marked with green squares. For

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instance, with a node group of a SS and a SU, the passing area is reduced from a circular ring region of a SS (Figure 15 (top)) to an intersection of two circular ring regions of a SS and a SU (Figure 15 (bottom)).

B. Short-Range Communication Applications

Considering an environment with a short communication range (e.g., 4 meters) and given the settings for long-range communication environments, the system performance is evaluated by using two kinds of signals, RF signal and ultrasonic signal. The study in [38] suggests that 2 $\sim$ 500 microseconds may be a reasonable value for response delay measurement of a ultrasonic signal. In this set of experiment, the response delay of a ultrasonic signal is assumed to be 500 milliseconds.

This subsection describes the performance differences between the proposed system with cooperation and that without cooperation. For a system using a RF signal, Figure 16 shows that given the device parameter $\alpha_{mu} = 1$ and the clocking scale factor $z_{mu_pu} = 1$, the system with $\eta \approx 0.1$ in a non-cooperative manner and that with $\eta \approx 0.1$ in a cooperative manner achieve detection rate = 30% and detection rate = 50%, respectively. Comparing to the system performance (Figure 14) in an environment with long-range communications, the detection rate is more sensitive to the variation of the threshold value η_r (e.g., for $0 < \eta_r \leq 0.2$) due to the differences among the orders of system parameters, such as device delay, propagation delay, and timing resolution. With varying the threshold value η_r, the system with a RF signal has a convex degradation of detection rate as increasing η_r.

For a system using a ultrasonic signal, comparing to the system without cooperation, the proposed scheme with $\eta = 0.2$, $\alpha_{mu} = 1$, and $z_{mu_pu} = 1$ in a cooperative manner achieves about 18% improvement in detection rate (from 80% to 98%). Similar to the results in long-range communications, the cooperative scheme may be applied to significantly against the PUEA attack with proper settings of η_r.

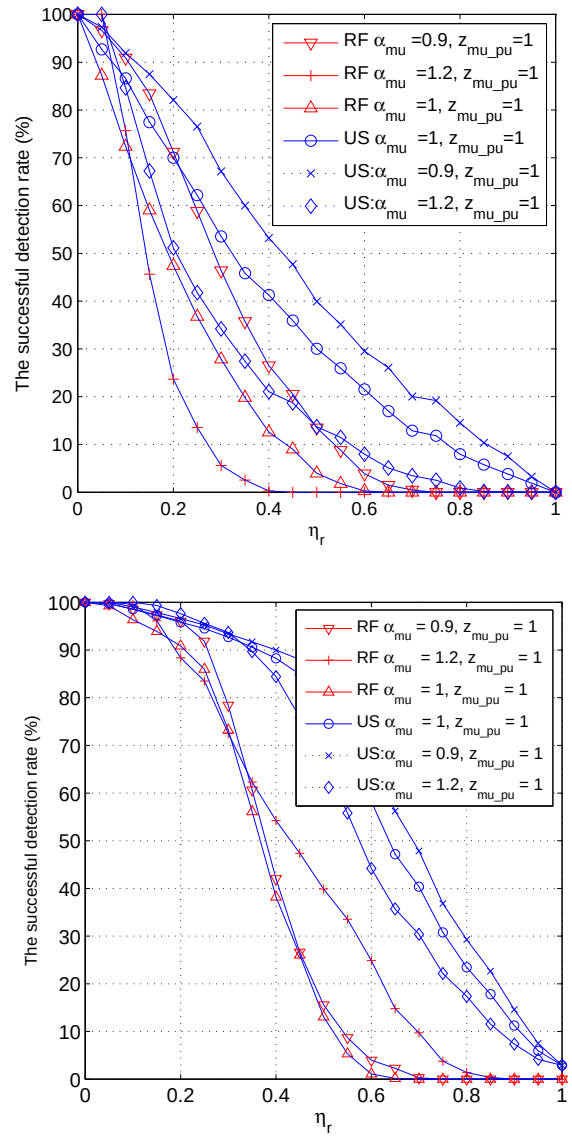

Fig. 16. The successful detection rate with varying η_r and the device delay of anti-nodes: non-cooperative method (top) and cooperative method with a SU (bottom).

C. Performance Comparison

The last set of experiments explore the detection performances of the proposed T^2DAP, the belief propagation-based (BP) scheme [10] and the energy detection-based (ED) scheme [11] with a RF signal. First, we consider the attack scenario described in Figure 10 in long-range communications (e.g., transmission range 30 meters). Given η_r and the device characteristics of a MU with respect to the PU, Figure 17 demonstrates the successful detection rate with varying the distance between a MU and the PU, d_{pu_mu}. As depicted in Figure 17, given $\eta_r = 0.1$, $\alpha_{mu} = 1$, $z_{mu_pu} = 1$ and with a small value of d_{pu_mu}, say $d_{pu_mu} = 1.5$ m, the successful detection rates of DBSS, BP, and ED are about 20%, 0%, and 1%, respectively. For a larger value of d_{pu_mu}, say $d_{pu_mu} =$ 7 m, the successful detection rates of T^2DAP, BP, and ED are about 100%, 37%, and 9%, respectively. Observe that the detection performance of T^2DAP is concave upward from $d_{pu_mu} = 0$ m to $d_{pu_mu} = 1.5$ m. This is because the differences of device characteristics and clocking information of a MU and the PU implicitly affect the successful detection rate.

For determining a proper range of the threshold value η_r, Figure 17 presents the detection performances of $\eta_r = 0.2$ and $\eta_r = 0.3$, respectively. The simulation results show that even with the worst case (i.e. $\alpha_{mu} = 1$ and $z_{mu_pu} = 1$), the detection system with η_r, which belongs to [0.1, 0.2], has a smaller blind detection range for $d_{pu_mu} \leq 2$ m and a better detection performance for $d_{pu_mu} > 2$ m. Therefore, η_r in [0.1, 0.2] may be a good choice of the detection threshold for the T^2DAP scheme. Accordingly, the proposed system with $\alpha_{mu} = 1$ and $z_{mu_pu} = 1$, and $\eta_r = 0.1$ has high sensitivity to d_{pu_mu} and is significantly superior to BP scheme and ED scheme from $d_{pu_mu} = 1.5$ m to $d_{pu_mu} = 10$ m.

In order to better assess the detection quality, the relation-

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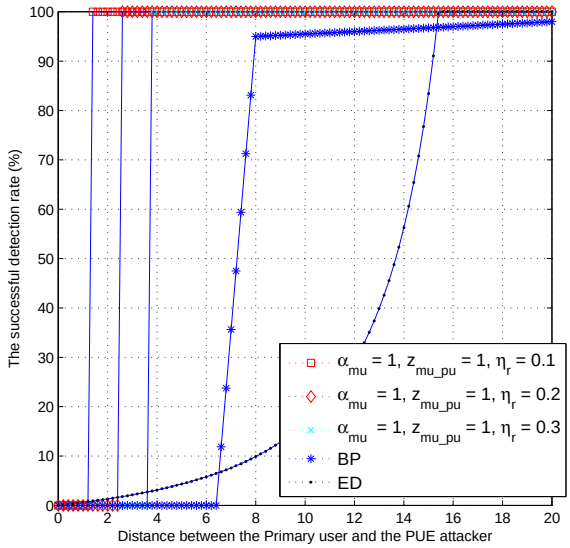

Fig. 17. Performance comparison (long range): the successful detection rate with varying the distance between a MU and the PU

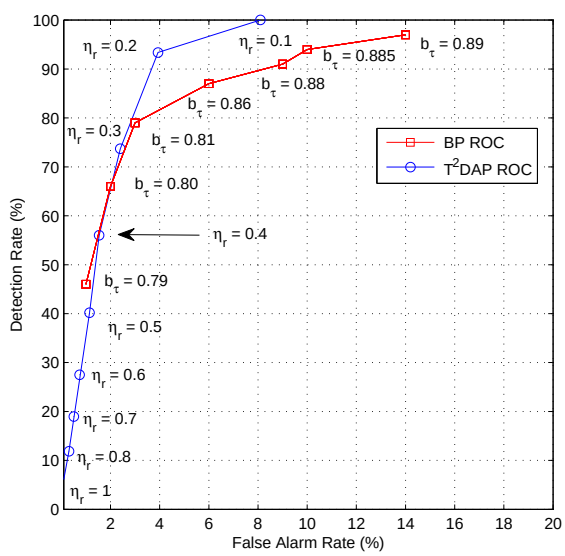

Fig. 18. The detection quality: $\mathrm{T^2DAP}$ with trust threshold η_r vs. BP with belief threshold b_τ.

ship between false alarm rate and setting of the threshold value is explored. As expected, Figure 18 shows that a smaller threshold trust value of η_r leads to a higher detection rate and a higher false alarm rate. Comparing to the ROC curve of BP [10] with a given false rate (say 5%), the detection rate of $\mathrm{T^2DAP}$ with η_r in [0.1, 0.2] ($\approx 95\%$) is better than that of BP ($\approx 85\%$). As illustrated in Figure 18, in order to perform the proposed detection and identification mechanism, the appropriate value of η_r may be within the range (0.1-0.2).

VI. Conclusion

This paper develops a new scheme for sensor tasking (e.g. a novel scheme for sensing decision and selecting sensing sensors) in CRSNs, provides the conceptual principle of the proposed scheme against the PUEA attack, and details its implementation. With the development of joint spectrum allocation and topology control algorithms, the performance of real-time sensing applications may be further improved. That is, controlling channel access and end-to-end delay for real-time surveillance applications are achievable.

Although the proposed strategy may achieve effective attack detection, further experimental and theoretical extensions are possible. In future work, we hope to improve the reliability and practicality of the system by considering the impact of node mobility on detection performance, developing feasible algorithms for sensor scheduling and power management, searching for appropriate cooperative schemes, and exploring intelligent physical layer security mechanisms.

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